MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6943

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, sharp cutoff, subminature
Outline --- 3-11 (EIA)
Base --- E8-10 (long leads)
Envelope --- T3
Cathode --- Coated unipotential
Base connections:

Pin No. ---	1	2	3	4	5	6	7	8
Element ---	g1	k, sd	h	g3	a	h	g2	k, sd

Failure rate assurance: See note 1

	AFR %/200 hours	RFR %/200 hours
Failure rate class (1) - Inoperatives	1.3	4.6
Failure rate class (2) - All end points	5.4	11.0
Failure rate class (3) - All end points	2.7	5.5

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	eb	Ec1	Ec2	Ec3	Ehk	Rk	t
Unit:	V	Vdc	v	Vdc	Vdc	Vdc	v	Ohms	sec
Maximum:	6.9	250	360	0, -55	150	0	200	---	---
Minimum:	5.5	---	---	---	---	---	---	---	---
Maximum for accelerated warmup:	8.5	---	---	---	---	---	150	---	10
TEST CONDITIONS:	6.3	100	---	0	100	0	0	150	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Cycles	Rg1	Ik	Pp	Pg2	TE	Alt	Radiation	
Unit:	---	Meg	mAdc	W	W	°C	ft	NVT	NV
Maximum:	---	1.0	15.0	1.0	0.33	+250	Note 12	10^{16}	10^{12}
Minimum:	---	---	---	---	---	---	---	(see note 2)	
Maximum for accelerated warmup:	50	---	---	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS BOGEY	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current		0.4	II	If	165	175	185	mA
1336	Heater-cathode leakage		0.4	II	Ihk	---	---	15.0	µAdc
1226	Total grid current	Eb = Ec2 = 125 Vdc; Ec1 = -2.5 Vdc; Rk = 0; Rg1 = 1.0 Meg	0.4	II	Ic1	0	---	-0.1	µAdc
1256	Electrode current (1) (anode)		0.4	II	Ib	6.0	8.0	10.0	mAdc
1256	Electrode current (2) (anode)	Ec1 = -8.5 Vdc; Rk = 0	0.4	II	Ib	---	---	100	µAdc
1256	Electrode current (screen)		0.4	II	Ic2	---	2.3	3.6	mAdc
1306	Transconductance (1)		0.4	II	Sm	2,800	3,600	4,400	µmhos
1201	Short and discontinity detection		0.4	II	---	---	---	---	---
---	Suppressor	See note 11	0.4	II	---	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	E(g1 - all) = -100 V	2.5	I	Rg1-all	500	---	---	Meg
		E(p - all) = -300 V			Rp-all	500	---	---	Meg
		E(g2 - all) = -200 V			Rg2-all	500	---	---	Meg
1306	Transconductance (2)	Ef = 5.5 V	2.5	I	ΔSm Ef	---	---	15	%
1256	Electrode current (3) (anode)	Ec1 = -5.5 Vdc; Rk = 0	2.5	I	Ib	5.0	---	---	µAdc
1266	Grid emission	Ef = 7.5 V; Rg1 = 1.0 Meg; Ec1 = -8.5 Vdc; Rk = 0 (see note 3)	2.5	Code G	Ic1	0	---	-0.5	µAdc
1246	Audio frequency noise	Esig = 70 mVac; Ec2 = 19 Vdc; Rg1 = 0.1 Meg; Rg2 = 1,000 ohms; Ck = 1,000 µF; Rp = 0.2 Meg (see note 8)	2.5	I	---	---	---	---	---
---	Hum	Ef = 6.3 at 400 Hz; Eb = Ec1 = Ec2 = 0; Rk = 10,000 ohms (see note 4)	2.5	I	Hum	---	---	15	mvpp
---	Operation time		6.5	Code F	t	---	---	20	sec
1311	Anode resistance		6.5	Code F	rp	0.2	0.3	---	Meg
1331	Direct-interelectrode capacitance	Shield 318	6.5	Code F	Cgp	---	---	0.02	pF
					Cin	2.2	---	3.8	pF
					Cout	2.2	---	3.8	pF

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS			UNIT
						MIN	BOGEY	MAX	
	Quality conformance inspection, part 2 -Continued								
1031	High-frequency vibration	Rp = 10,000 ohms; Ck = 1,000 µF	1.0	I	Ep	---	---	40.0	mVac
---	White noise	Rp = 10,000 ohms; Ck = 1,000 µF; 15 G peak (see note 6)	2.5 2.5	I I	ep Ep	--- ---	--- ---	300 40	mvpp mV
1116	Lead fatigue		2.5	Code G	---	4	---	---	Arcs
1041	Shock	750 G; (hammer angle = 48°) (see note 7)	20	---	---	---	---	---	---
1031	Sweep-frequency vibration fatigue		6.5	See note 5	---	---	---	---	---
---	Post-shock and sweep-frequency vibration fatigue test end points:								
1031	High-frequency vibration		---	---	Ep	---	---	120	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	---	15	µAdc
1306	Transconductance (1) change of individual tubes		---	---	ΔSm	---	---	20	%
2126	Envelope strain		4.0	I	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---	---
	Quality conformance inspection, part 3								
1506	Heater-cycling life (1)	Ef = 7.0 V; 1 minute on, 4 minutes off; Ehk = 140 Vac; Eb = Ec1 = Ec2 = 0; 300 cycles	2.0	Code H	---	---	---	---	---
1506	Heater-cycling life (2)	Ef = 8.5 V; 10 seconds on, 4 minutes (min) off; 300 cycles; Ehk = +150 Vdc; Eb = Ec1 = Ec2 = 0	10.0	Code G	---	---	---	---	---
---	Heater-cycling life test (2) end points:								
1336	Heater-cathode leakage		---	---	Ihk	---	---	15	µAdc
1211	Insulation of electrodes		---	---	Rg1-all Rp-all Rg2-all	50 50 50	--- --- ---	--- --- ---	Meg Meg Meg

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS BOGEY	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 - Continued								
1516	Stability life (2 and 20 hours)	Ebb = 250 Vdc; Tie g2 to p; Rk = 470 ohms; Ehk = +200 Vdc; Rg1 = 1.0 Meg; Rp = 15,000 ohms; TA = room; Rhk = 5,000 ohms	1.0	Code H	---	---	---	---	---
---	Stability-life test end points:								
1306	Transconductance (1), change of individual tubes		---	---	ΔSm t	---	---	15	%

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	FAILURE RATE CLASS (2) LIMITS MIN	FAILURE RATE CLASS (2) LIMITS MAX	FAILURE RATE CLASS (3) LIMITS MIN	FAILURE RATE CLASS (3) LIMITS MAX	UNIT
1501	Intermittent life	Stability life test; TE - 250°C (minimum) (see notes 9 and 10)	---	---	---	---	---	---
---	Intermittent life-test end points: (200 hours)							
---	Inoperatives		---	---	---	---	---	---
1266	Grid current		Ic1	0	-0.5	0	-1.0	μAdc
1301	Heater current		If	165	190	160	195	mA
1306	Transconductance (1), change of individual tubes		ΔSm t	---	25	---	30	%
1306	Transconductance (2)		ΔSm Ef	---	25	---	30	%
1336	Heater-cathode leakage		Ihk	---	15	---	15	μAdc
1211	Insulation of electrodes		Rg1-all	20	---	10	---	Meg
			Rp-all	20	---	10	---	Meg
			Rg2-all	20	---	10	---	Meg
1306	Transconductance (1), average change		Avg ΔSm t	---	15	---	---	%
	Individual lot acceptance			Allowable defectives to failure rate class (2) limits				
	Inoperatives			2				
	Total defectives including inoperatives			5				
	Failure-rate acceptance			Allowable defectives				
	Failure-rate class (1)	Inoperatives		5				
	Failure-rate class (2)	Total defectives including inoperatives		16				
	Failure-rate class (3)	Total defectives including inoperatives		8				

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NOTES:

1. The failure rate assurance values are taken from the operating characteristic curve of the life-test sampling plan. The acceptable failure rate (AFR) is based on a 5 percent producer risk point. The rejectable failure rate (RFR) is based on a 10 percent consumer risk point. (These values are comparable to the AQL and LTPD values in regular sampling plans.) For reasonable economy of operation, the manufacturer will operate at the AFR value or better; however, if lots as poor as the RFR value are submitted for acceptance, they have only a 10 percent chance of acceptance.

The AFR and RFR ratings are based on the sampling plan for the five lot moving process average.

This specification contains three failure rate evaluations. First is one for inoperatives alone; second is an evaluation to the standard MIL electrical life-test end-point limits. The third failure rate is based on a modified set of end-point limits, which should be used if greater reliability is to be achieved.

2. The radiation ratings for this type shall be confirmed by a qualification test. The test shall be conducted in a suitable reactor with the required intensity and dosage no less than 10 percent of the specified rating. The tubes shall be measured both before and after irradiation for the electrical tests in quality conformance inspection, part 1. After irradiation, the tubes shall be within the limits specified in the tube specification sheet for the electrical tests in quality conformance inspection, part 1.

3. Prior to this test, tubes shall be preheated for 5 minutes at the conditions indicated below. Test within 3 seconds after preheating. The 3-minute test is not permitted. Grid emission is a destructive test so tubes subjected to it shall not be accepted under this specification, nor shall subsequent tests be performed on the same tubes.

Ef	Eb	Ec1	Ec2	Rk/k	Rg1/g1
V	Vdc	Vdc	Vdc	Ohms	Meg
7.5	250	0	150	1,000	1.0

4. Maximum total distortion of the filament supply voltage shall be 5 percent. The frequency response of the peak to-peak measuring device from 20 to 5,000 Hz shall be within 0.5 dB of its response at 400 Hz. Ground all leads except those for the cathode and heater lead No. 3. Measure hum voltage across specified Rk.

5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

6. The tube shall be rigidly mounted on a table vibrating so that the instantaneous values of acceleration shall constitute approximately a "WHITE NOISE" spectrum which is free from discontinuities from 100 to 5,000 Hz and such that the rms value of acceleration for frequencies outside this band shall constitute no more than 5 percent of the total rms acceleration. The spectrum of instantaneous acceleration shall be such that each octave of bandwidth delivers 2.3 + 0.2 G's rms acceleration. With this the case, the rms value of acceleration for any bandwidth within the specified spectrum is equal to:

$$\text{GRM Grms} = 2.3\sqrt{3.32 \log_{10}(f2/f1)}$$

where f2 and f1 are the upper and lower frequencies respectively of the band under consideration. The degree of clipping of the peak accelerations shall be such that the peak value of acceleration is at least 15 G's. Half the tubes in the sample shall be vibrated in position X, the other half in position Y. The voltage (ep) produced across the resistor (Rp) as a result of vibration shall be coupled through a compensating amplifier to a low-pass filter. The compensating amplifier shall have a high-input impedance (250 kohms or more) and shall be adjusted to compensate for any insertion losses in the filter. The combined frequency response of amplifier and filter shall be flat within ±0.5 dB from 50 to 8,000 Hz, shall be down no more than 5 dB at 10,000 Hz and at 20 Hz, and down at least 30 dB at 13,000 Hz. For reading the peak-to-peak value of output voltage, the filter output shall be fed directly to the input of a Ballantine Model 305 peak-to-peak electronic voltmeter, or equal, while the rms value shall be measured with the Hewlett Packard Model 400 C, or equal. The impedance of the anode and screen voltage supplies shall not exceed that of a 40 μF capacitor at 10 Hz.

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7. A grid resistor of 0.1 Meg shall be added; however, this resistor shall not be used when a thyratron type short indicator is used.

8. The rejection level shall be the VU meter reading obtained during calibration.

9. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

10. Intermittent life-test evaluation. The life test conducted in accordance with this specification shall be evaluated in two separate procedures. The first shall be an evaluation on an individual lot basis in a similar manner to the one normally specified in reliable tubes. The second evaluation shall be the determination of certain failure rates and their compliance to this specification. The life test shall be conducted in accordance with method 1501 except that the sampling plan will be changed from the presently specified double sampling plan to a single sampling plan with n = 40. Individual lot acceptance shall be based on this sample size and the conformance of the lot to the maximum allowable defectives for inoperatives and also for combined defectives as specified.

Failure rate control.

a. Purpose. It is the purpose of this specification to establish a means by which the military services may monitor and evaluate the failure rate of this tube type in order to provide a statistically valid description of this parameter that may be utilized in computing the reliability of equipments in which this type is used.

b. Description. The failure rates on this type shall be evaluated using the life-test information gathered on individual lots. Only those consecutive lots that pass the life-test criteria for individual lots shall be used in the calculations for failure rates. The failure rates shall be based on five lot moving sums with the cumulative number of failures for each failure rate class plotted on their respective failure rate charts (see figures 1, 2, and 3). Nonconformance of a tube to the 200-hour end points shall be considered a failure.

The failure rate charts as shown on figures 1, 2, and 3 consist of a plot of consecutive lots manufactured with each point indicating the cumulative number of failures in the last five lots as evaluated against the specified criteria. In the operation of this procedure, each time a new lot is added, the number of defectives from the earliest lot is dropped from the accumulation of defectives so that each plotted point on the chart will represent a total sample size of 200 tubes. Maverick samples and samples containing a "concentration of defects" are defined as follows and shall not be included in the cumulated regular samples.

Concentration of defects. A concentration of defects is defined as the presence of an excessive number of a single type of defect in the sample when it is ascertained that the reason for repetition of the same defect can be traced to one assignable cause, and the same defect repeats itself in all of the failing tubes of the sample that have been subjected to the same production or process conditions.

Maverick samples. Maverick samples are defined as samples containing an excessive number of defects which are not explained by a concentration of defects and which are presumed to be abnormal, for example, beyond 2.5 Sigma limits based on the AFR value representing the process average.

c. Qualification. In order to become a qualified source to this specification, it is mandatory that the manufacturer supply data on charts 1, 2, and 3 indicating that the failure rate is within the acceptable limits. A total quantity of 200 tubes life tested for 200 hours is required and this should be representative of two or more consecutively manufactured lots (for example, 100 tubes from each of 2 lots or 40 from each of 5 lots). In the event more than 40 tubes per lot are used in the life-test sample in order to accelerate qualification for failure rate, only the first 40 tubes life tested per lot shall be evaluated for the individual lot acceptance criteria.

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d. Maintenance of failure rate. When the manufacturer has become a qualified source to this specification and has received proper approval from the Naval Electronic Systems Command, those lots utilized for qualification and subsequent lots where the five lot moving sum failure rate continues within the specification, are to be considered acceptable to this specification and may be marked and shipped accordingly.

e. Nonconformance of failure rate. The failure rate will be considered nonconforming when the total number of defectives (for the particular failure rate under evaluation) accumulated from the past five lot life test exceeds the number permissible as contained in the specification and on the failure rate chart. The lot of tubes which caused the failure rate limit to be violated shall be considered acceptable if it conforms to the individual lot failure criteria, and can be marked and shipped in accordance with this specification. In the event two or more lots of the last five submitted for life test fail the criteria for individual lot release, the failure rate control shall also be considered nonconforming. When a failure rate criteria has been violated, the manufacturer has lost qualification to this specification and cannot mark and ship any subsequent lots to this specification without being requalified.

f. Requalification for failure rate. Since all three failure rates specified in this specification are computed from the same life-test data, a violation of one failure rate requires requalification for all three failure rates. Requalification shall follow the same procedure as for original qualification. Starting after the last lot which rendered disqualification, a life test sample of 200 tubes from the next two new consecutive lots will be required to conform to this specification before qualification is reinstated to the manufacturer.

11. Reject for open suppressor if anode current does not decrease by a minimum of 10 percent when Ec3 is changed from 0 to -100 Vdc.

12. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.

13. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

User activities:
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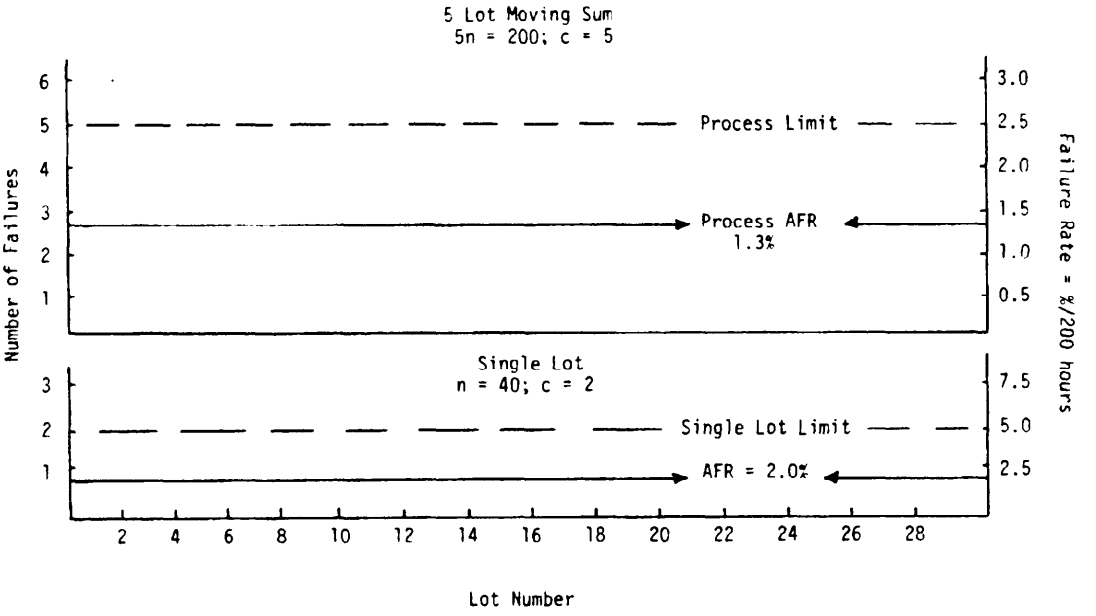


FIGURE 1. Failure rate class 1 (inoperatives) chart.

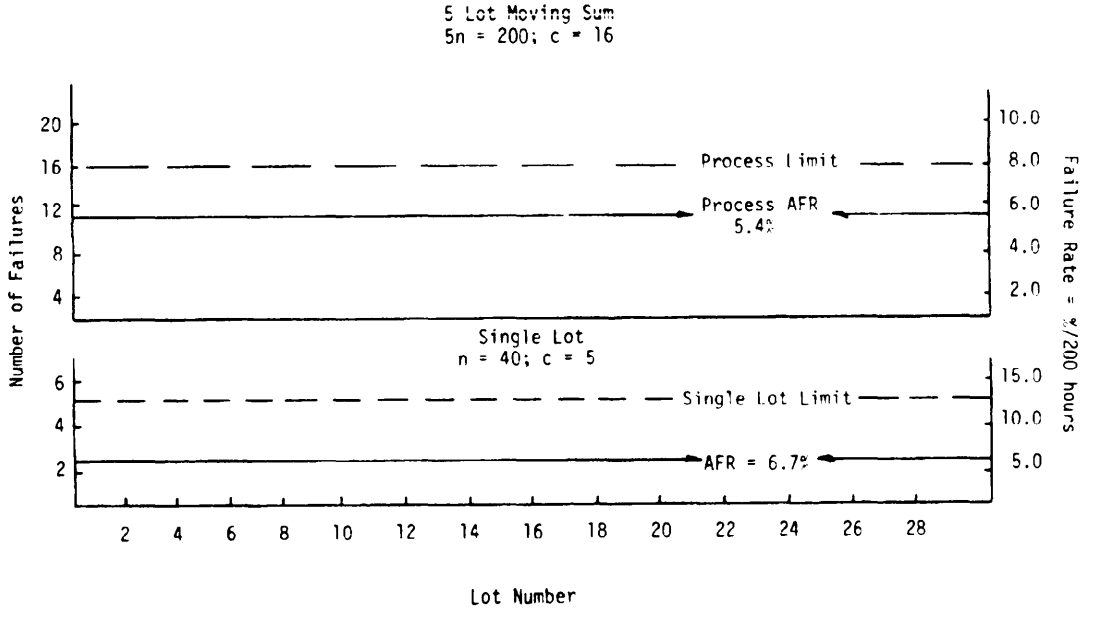


FIGURE 2. Failure rate class 2 chart.

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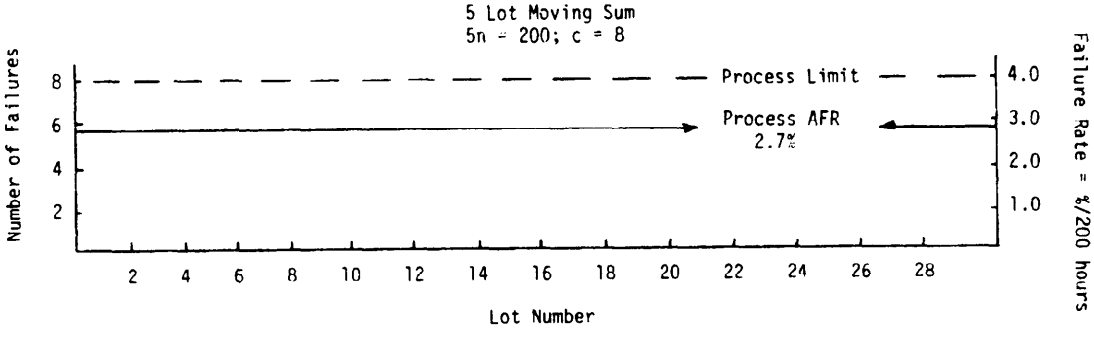


FIGURE 3. Failure rate class 3 chart.

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